

Cooling Water System — Interlock & Isolation Procedure

Lockout/Tagout Sequence LOTO-7: Valve V-104 & Pump P-201A

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1. Purpose

Gate Valve **V-104** and Centrifugal Pump **P-201A** are functionally interdependent: V-104 is the sole suction isolation point for P-201A, and P-201A's running state determines whether V-104 can be safely opened for service. This procedure defines the mandatory isolation sequence between the two assets before either may be serviced.

2. Isolation Sequence — Servicing P-201A

Before performing any maintenance on Pump P-201A, Valve V-104 shall first be closed and locked out to isolate the pump suction. Disassembly of P-201A shall not proceed until the V-104 lockout tag is confirmed in place and zero-energy state is verified at the pump casing.

3. Isolation Sequence — Servicing V-104

Before performing any maintenance on Valve V-104, Pump P-201A shall first be de-energized at the motor control center and locked out. V-104 shall not be opened for service while P-201A remains electrically energized, regardless of confirmed pump-stopped status on the control room display.

4. Electrical Interlock

V-104 and P-201A are electrically interlocked at the control system level: P-201A is prevented from starting unless V-104's position switch confirms the valve is fully open. This interlock is a protective control and does not replace the physical lockout steps in Sections 2 and 3.

5. Compliance References

All isolation activities under this procedure, for both V-104 and P-201A, shall comply with OSHA Standard 1910.147, The Control of Hazardous Energy (Lockout/Tagout).